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Data-Driven Analysis of the Impact of Weather & Traffic Conditions on Food Delivery Time

Statistical analysis for a more realistic delivery time estimates

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PROJECT BACKGROUND

Business Problems

The estimated delivery time shown to customers is generally the same under all conditions, even though weather and traffic conditions in the field can vary considerably. This project uses historical data to determine which factors actually have an impact, rather than relying on assumptions.

1

ETA is not adaptive to actual conditions

Estimated time has not yet been adjusted for actual weather and traffic conditions

2

Facts are not yet clear

The operational factors that truly have an impact have not yet been definitively identified.

3

The impact of onfield conditions has not yet quantified

There has not yet been a quantification of the combined impact of adverse conditions.

4

Risk of an increase in complaints received from customers

The gap between expectations and reality might cause complaints from customers

DATA

Dataset Used

1.000	9	4	11,7%
Total order	Total column	Total missing columns	Affected rows

Columns in the dataset

- Order_ID, Distance_km, Preparation_Time_min
- Weather, Traffic_Level, Time_of_Day
- Vehicle_Type, Courier_Experience_yrs
- Delivery_Time_min (target)

Analysis target: explaining variations in Delivery_Time_min using a combination of operational factors (weather, traffic, etc) and distance/preparation factors, followed by a statistical significance test.

DATA

Data Cleaning

Imputation Strategy

Categorical: Weather, Traffic_Level, Time_of_Day → handled with mode

Numerical: Courier_Experience_yrs → handled with median

Missing Value Pattern

- 117 unique rows affected out of a total of 1,000 (11.7%)
- 114 rows are blank in only 1 column
- Only 3 rows are blank in more than 1 column

Why not just delete it?

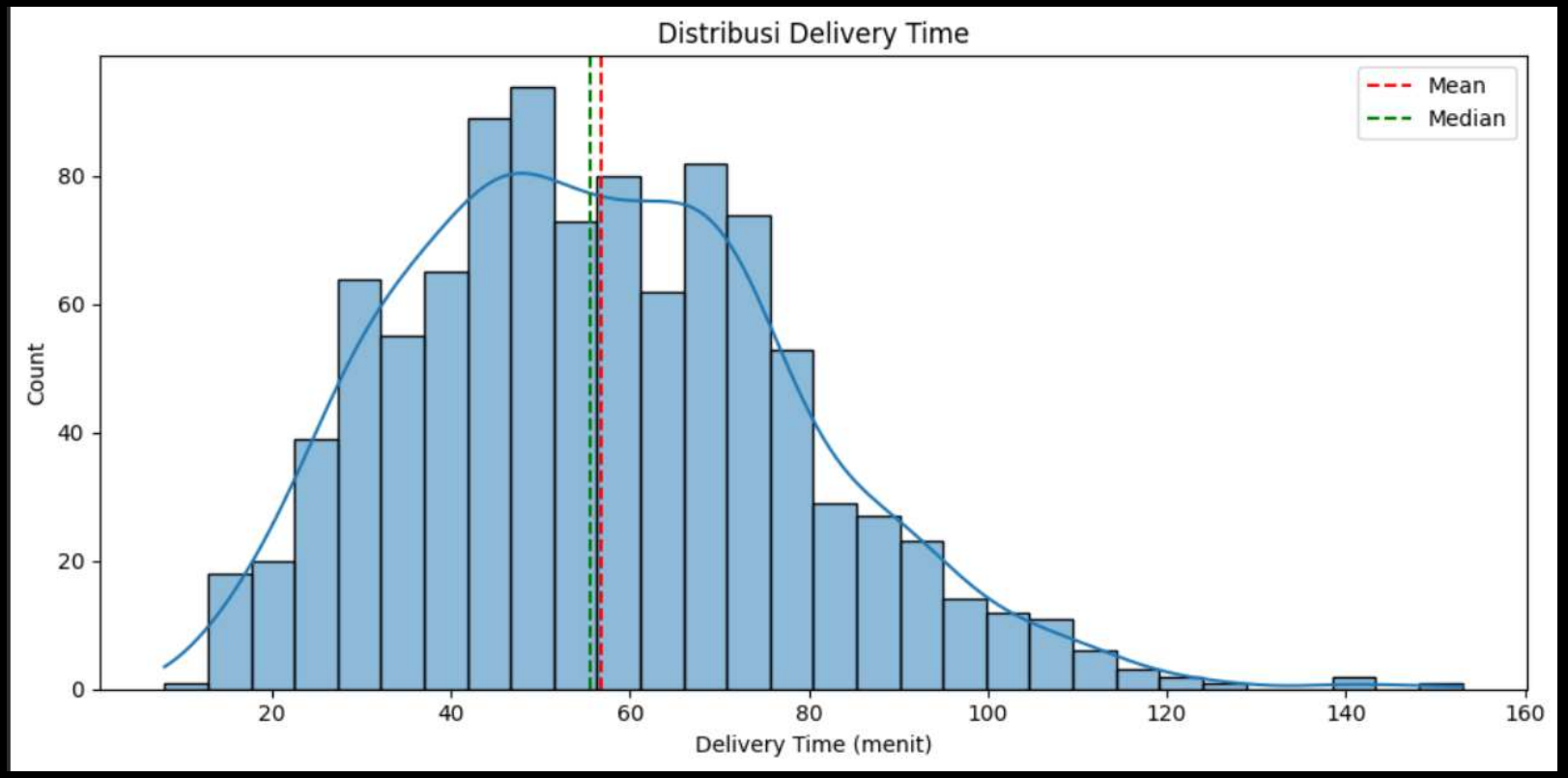
The missing data are randomly scattered across 117 rows (rather than concentrated in one place), and only 3 rows have missing values in more than one column.

Deleting them would mean discarding 11.7% of still-useful data, so imputation was chosen as the safer approach to preserve the sample size for subsequent statistical tests.

	dtype	Missing	Duplicated	Unique
Order_ID	int64	0	0	1000
Distance_km	float64	0	0	785
Weather	object	30	0	5
Traffic_Level	object	30	0	3
Time_of_Day	object	30	0	4
Vehicle_Type	object	0	0	3
Preparation_Time_min	int64	0	0	25
Courier_Experience_yrs	float64	30	0	10
Delivery_Time_min	int64	0	0	108

DATA EXPLORATION

Distribution of the Delivery Time Data



56,7

Average (minutes)

55,5

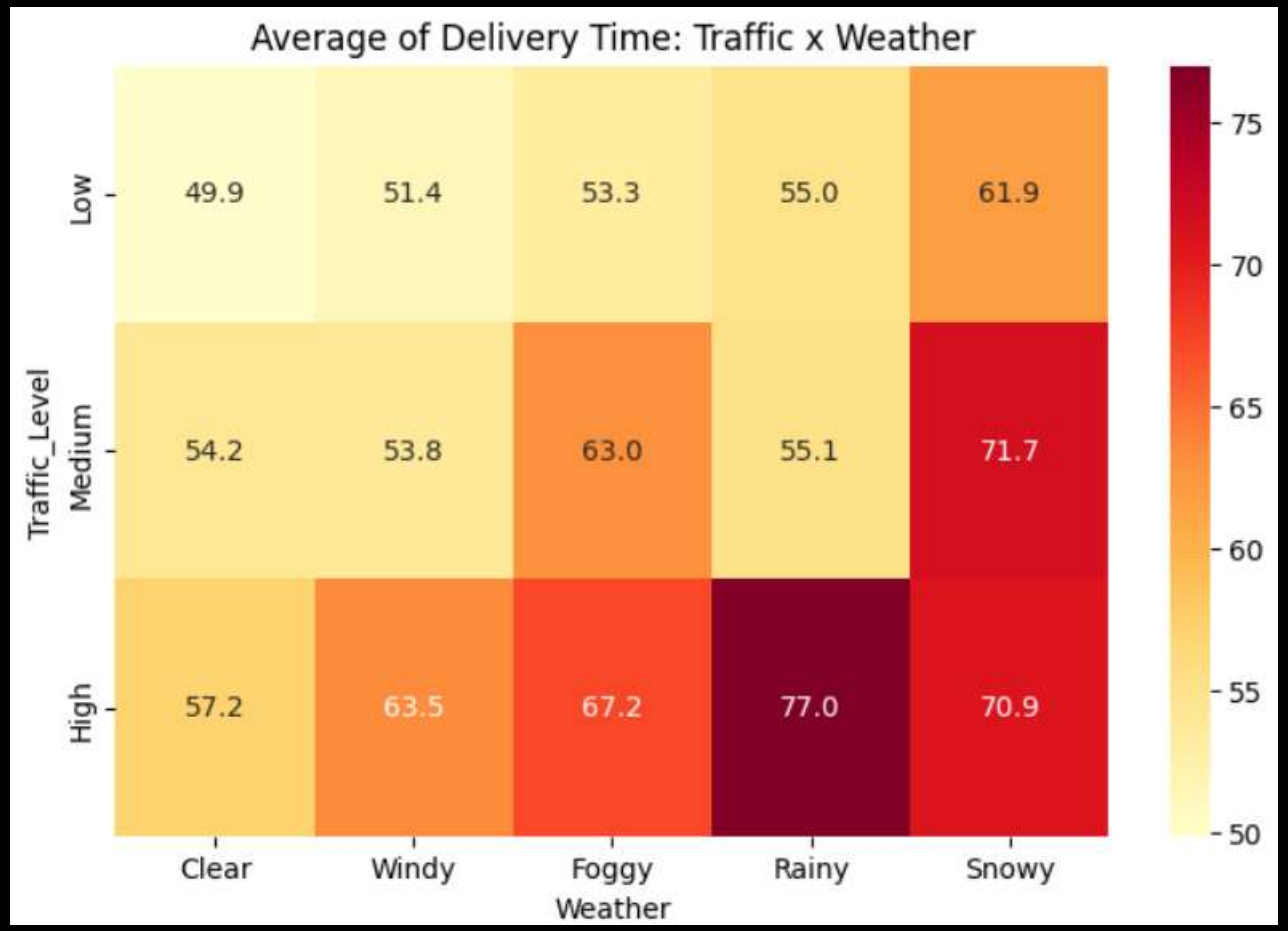
Median (minutes)

0,51

Skewness (right-skewed), a small number of orders took much longer than average

DATA EXPLORATION

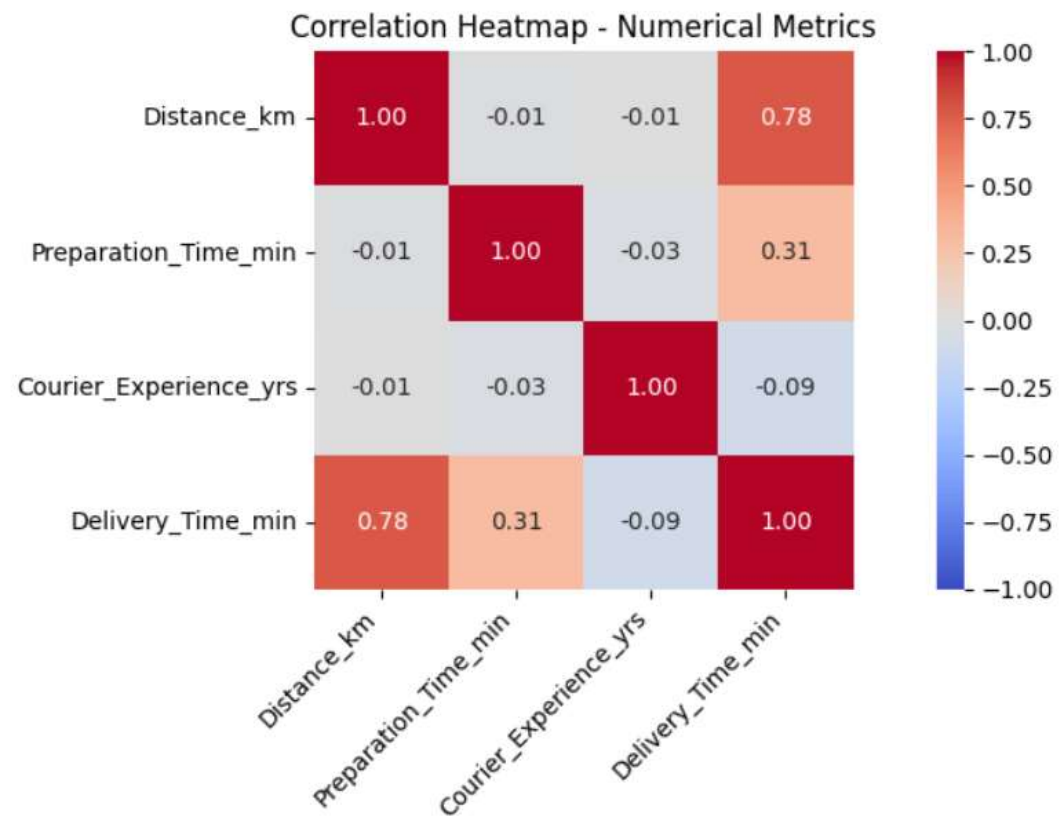
Weather & Traffic Slow Down Deliveries



Snowy weather affects delivery time the most

Snowy weather conditions consistently have the longest delivery times, with the longest delivery occurring during rainy weather and heavy traffic conditions,

Correlation Between Numerical Metrics



Distance is the dominant predictor of order delivery time

This is followed by the order preparation time at the restaurant/eatery itself, which has a moderate effect (r = 0.31). The courier's experience has virtually no effect, with a correlation level of only (r = −0.09). These three predictors are also not correlated with one another, so the interpretation of each one's effect on delivery time can be considered independent.

DATA EXPLORATION

Combination of Poor Conditions = Worst Delay

Best Condition

49,9

minutes

Low Traffic + Clear Weather

+54%

slower

Worst Condition

77

minutes

High Traffic + Rainy Weather

Interaction effects: The combined impact of high traffic and rain is worse than if the two effects were simply added separately (simple estimate: ~68 minutes; actual: 77 minutes). Snowy weather, meanwhile, remains consistently severe at all traffic levels (average: 68.2 minutes), whereas Rainy weather only rises sharply when traffic is high. Travel distance remains strongly correlated ($r = 0.78$) across all conditions.

DATA EXPLORATION

Relationship Between Distance and Delivery Time



r = 0,78

Strong correlation between distance and delivery time

The greater the distance travelled, the longer the delivery time; this pattern is consistent across all weather and traffic conditions.

METHODS

From Visual Patterns to Statistical Evidence

1

Normality Check

Shapiro-Wilk

p = 0,0000409

Not normal

2

Kruskal-Wallis

H-test

p < 0,00001

Significant

3

Effect Size

Epsilon-squared

ε^2 = 3,3% & 3,5%

Small impact on delivery time

4

Insight

& Recommendations

4 Business

recommendations

DELIVERY TIME PERFORMANCE DASHBOARD

Weather
- Clear
- Foggy
- Rainy
- Snowy
- Windy

Traffic_Level
- High
- Low
- Medium

Vehicle_Type
- Bike
- Car
- Scooter

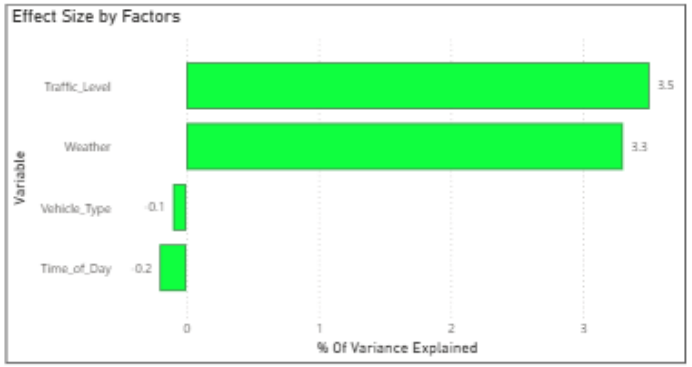
Time_of_Day
- Afternoon
- Evening
- Morning
- Night

Total Order
1K

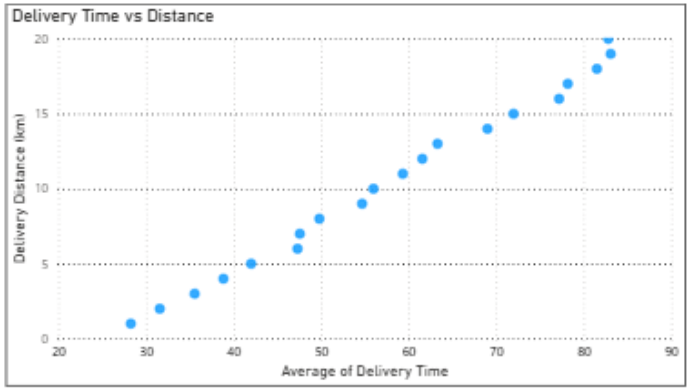
Average Distance (km)
10.06

Average Delivery Time (min)
56.73

Median of Delivery Time (min)
56







Delivery Time by Traffic Level x Weather Condition

Traffic_Level	Clear	Foggy	Rainy	Snowy	Windy	Total
High	57.16	67.24	77.05	70.90	63.48	**64.81**
Medium	54.24	63.03	55.06	71.73	53.79	**56.45**
Total	**53.15**	**59.47**	**59.79**	**67.11**	**55.46**	**56.73**

Summary

Variabel	Sum of H-statistic	Sum of epsilon squared	Sum of % variance explained
Traffic_Level	37.07	0.04	3.50
Weather	37.05	0.03	3.30
Vehicle_Type	0.99	0.00	-0.10
Time_of_Day	0.94	0.00	-0.20
Total	**76.05**	**0.07**	**6.50**

RESULTS

Kruskal–Wallis Test Results

Testing whether the average delivery-time ranks differ significantly between groups for each categorical variable (α = 0.05).

Variable	Number of Groups (k)	H-statistic	p-value	Result
Weather	5	37,05	< 0,00001	Significant
Traffic_Level	3	37,07	< 0,00001	Significant
Vehicle_Type	3	0,99	0,611	Not significant
Time_of_Day	4	0,94	0,816	Not significant

Conclusion: Only Weather and Traffic_Level have been shown to have a significant effect on delivery time. There is insufficient evidence to conclude that Vehicle_Type and Time_of_Day have an effect.

RESULTS

Effect Size: Just How Powerful Its Effect on Delivery Time?

The p-value only indicates whether an effect exists. The effect size (epsilon-squared) indicates how substantial that effect is on the variable being analysed, calculated from the preceding Kruskal–Wallis results.

$$\varepsilon^2 = (H - k + 1) / (n - k)$$

Variables	ε^2	% variance	Interpretation
Traffic_Level	0,0352	3,5%	Weak
Weather	0,0332	3,3%	Weak
Vehicle_Type	≈ 0	≈ 0%	Very Weak
Time_of_Day	≈ 0	≈ 0%	Very Weak

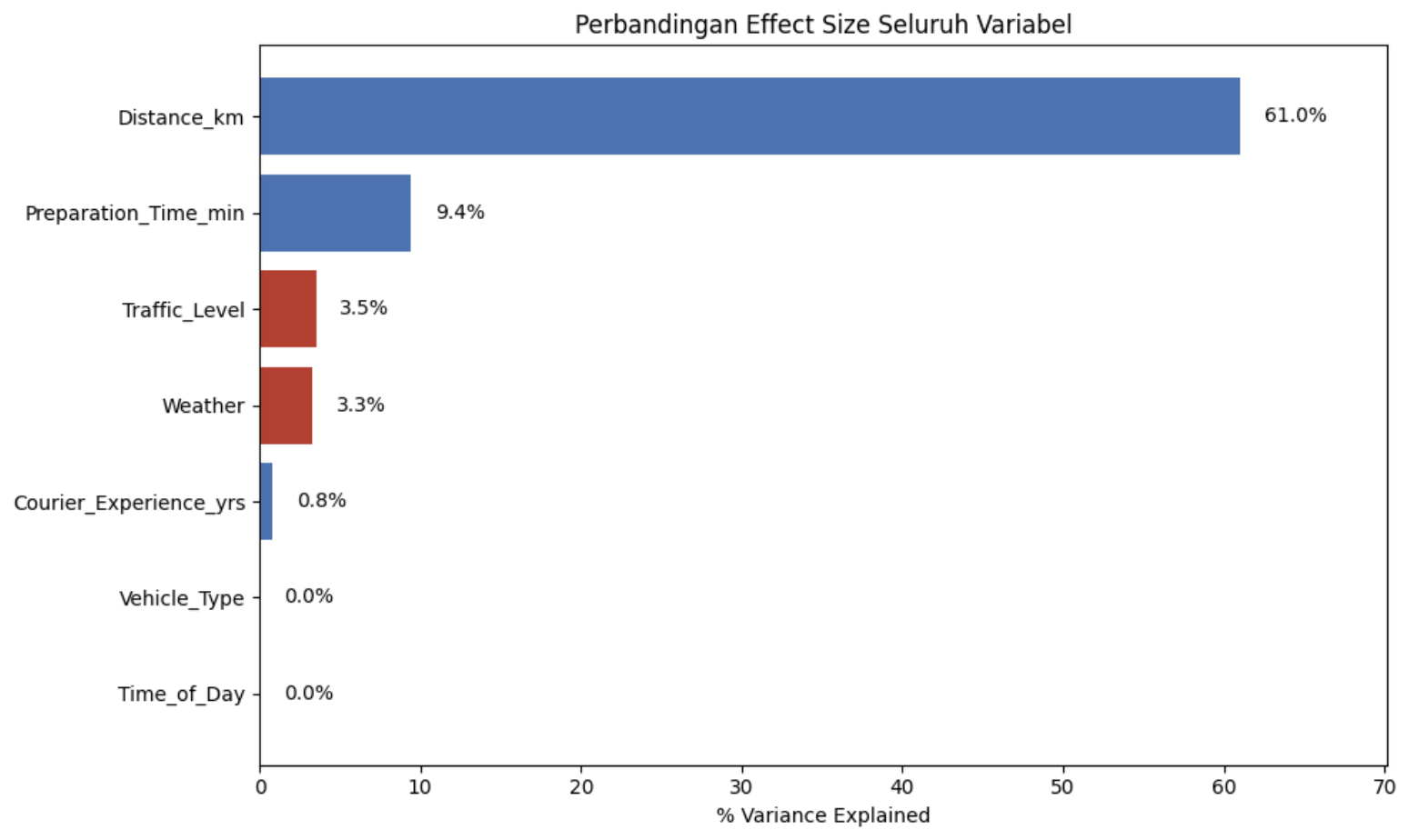
Scales of Interpretation

< 0,01	Very Weak
0,01 – 0,08	Weak
0,08 – 0,26	Medium
> 0,26	Powerful

Traffic_Level and Weather are statistically significant, but their magnitude falls into the small category. A real signal, not a dominant factor.

RESULTS

Travel Distance Remains the Dominant Factor



$r^2 = 0,61$

The variation in delivery times is explained solely by travel distance.

Weather (0,033) and traffic (0,035) were shown to be statistically significant ($p < 0,00001$), but their effect size was considerably smaller than that of travel distance.

CONCLUSION

Insights & Business Recommendations

1

Adaptive ETA by Onfield Condition

Use the historical average table for each traffic and weather combination as the basis for estimates, rather than using one figure for all conditions. The 54% gap between the best and worst conditions is too large to ignore.

2

Focus on Travel Distance, With Weather and Traffic Conditions as its Main Buffer

Prioritise distance as the primary basis for estimation (61% of variance explained), with weather & traffic conditions as its main buffer (~3% of variance explained); stop optimising based on vehicle type or order time, as neither has been proven to have an effect.

3

Differentiate Time Buffers

Consistent buffer for Snowy weather across all traffic conditions; a specific buffer for Rainy weather during high traffic, following the pattern of interaction effects identified.

4

Make a Record of ETA vs Actual Delivery Time

Begin collecting the ETAs shown to customers to measure the effectiveness of future recommendations and validate their impact on complaints.

Thank You